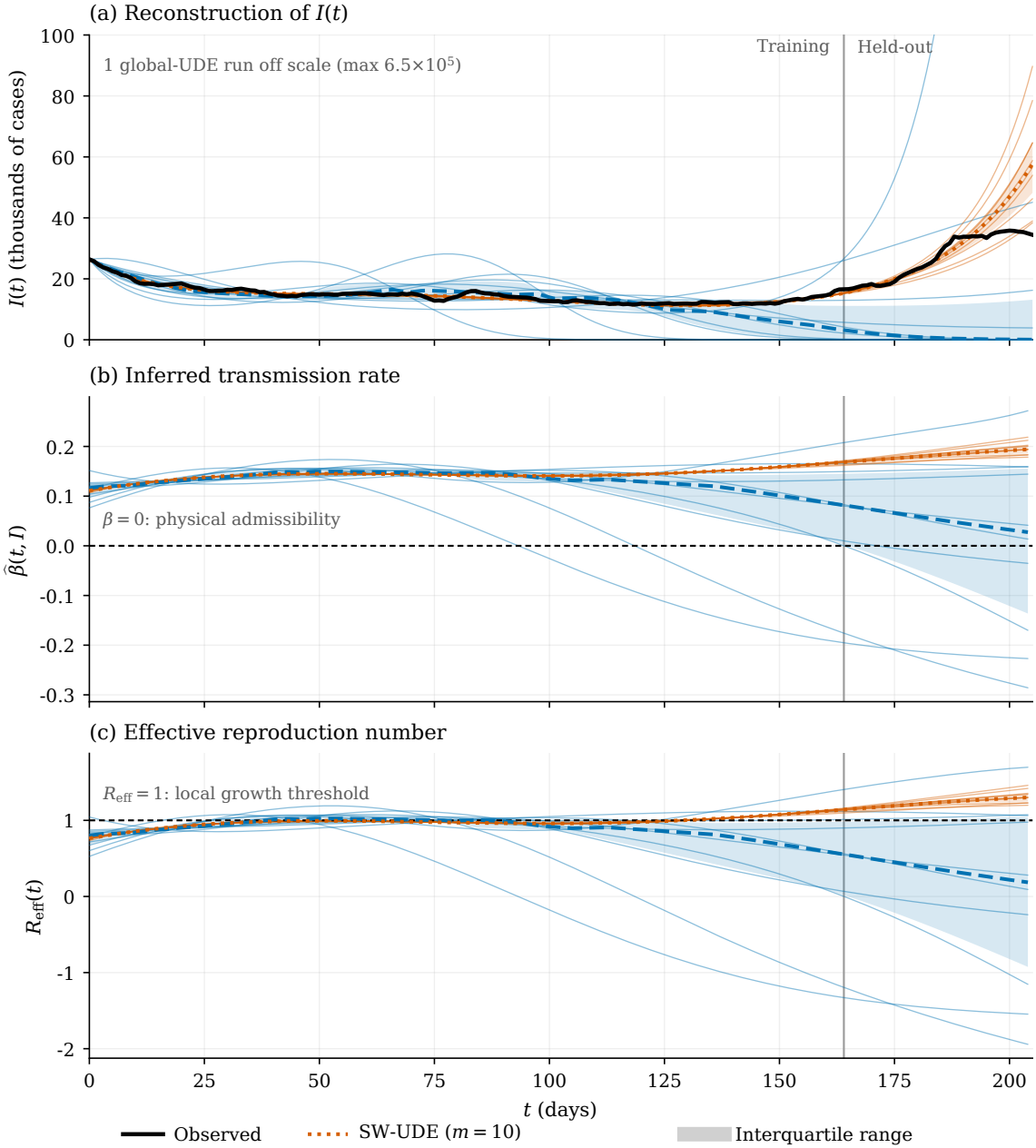

(a) Reconstruction of $I(t)$
1 global-UDE run off scale (max 6.5×10^5)
Training
Held-out
$I(t)$ (thousands of cases)
100
80
60
40
20
0
(b) Inferred transmission rate
$\beta = 0$: physical admissibility
$\hat{\beta}(t, I)$
0.2
0.1
0.0
-0.1
-0.2
-0.3
(c) Effective reproduction number
$R_{\text{eff}} = 1$: local growth threshold
$R_{\text{eff}}(t)$
1
0
-1
-2
0
25
50
75
100
125
150
175
200
t (days)
Observed
Global UDE
SW-UDE ($m = 10$)
Individual initializations ($n = 10$)
Interquartile range